NSO8062 Geofüüsikaline andmetöötlus

Loeng 3: NetCDF failid

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19.02.2026

Ajakava ja teemad

Nädal	Teema	Õppejõud
1	Sissejuhatus ainesse	S-T. Luik
2	Erinevad andmete tüübid ja lihtsamad tegevused nendega.	S-T. Luik
3	NetCDF failide loomine, muutmine jne.	S-T. Luik
4	Keri andmed. Profiilide visualiseerimine, andmete eemaldamine, silumine, gradientide leidmine.	S-T. Luik
5	Andmete 1D interpoleerimine ja korrelatsioonid, statistikud jne.	S-T. Luik
6	Tuuleandmed ja nende visualiseerimine	S-T. Luik
7	2D andmed, interpoleerimine, visualiseerimine (nt kaart)	
8	TBD	
9	TBD	
10	TBD	
11	TBD	
12	TBD	
13	TBD	
14	TBD	
15	Konsultatsioon	
16	**Eksam**	
17		
18		
19	10/06/2026 kevadsemestri eksamite sooritamise lõpptähtaeg	
20	15/06/2026 kevadsemestri lõpp	

NetCDF (Network Common Data Form)

Format commonly used for storing multi-dimensional scientific data, such as climate data, oceanographic data, and more. It is particularly useful for large datasets because it allows efficient access and sharing of data.

Key Features of NetCDF:

- Multi-Dimensional Data: NetCDF can handle data with multiple dimensions (e.g., time, latitude, longitude).
- Self-Describing: It includes metadata (data about data), which helps describe the contents and structure of the datasets.
- Platform Independence: NetCDF files can be read and written on different platforms without needing conversion.
- Compression: It supports data compression to save storage space.

Structure of a NetCDF File

NetCDF file typically consists of the following components:

- **Dimensions**: Define the size of each axis of the data (e.g., time, latitude, longitude).
- **Variables**: The actual data arrays that are stored, which can be multi-dimensional.
- **Attributes**: Metadata associated with dimensions and variables, providing context (e.g., units, descriptions).
- **Global Attributes**: Information about the entire dataset (e.g., title, source, history).

- EXAMPLE: series_bot_lumi_057_a0.nc

1x1 struct with 8 fields

Field	Value
Filename	'C:\Users\Stella\OneDrive -
Name	'/'
Dimensions	1x4 struct
Variables	1x5 struct
Attributes	[ ]
Groups	[ ]
Format	'netcdf4_classic'
Datatypes	[ ]

Working with NetCDF Files in MATLAB

Common Functions:

- Reading Data:

  ncinfo: Retrieves information about the contents of a NetCDF file.

  ncdisp: Display the contents of the netCDF file.

  ncread: Reads data from a specified variable in the NetCDF file.

- Writing Data:

  ncwrite: Writes data to a variable in a NetCDF file.

  ncwriteatt: Writes a global attribute to a NetCDF file.

Creating a NetCDF file in MATLAB

This creates a variable named temperature with three dimensions: time, latitude, and longitude:

```
nccreate('example.nc', 'temperature', 'Dimensions', {'time', 10, 'latitude', 5, 'longitude', 5});
```

You can specify the data type of the variable using the 'Datatype' option. Common types include 'double', 'single', 'int32', 'int16', 'char', etc:

```
nccreate('example.nc', 'temperature', 'Dimensions', {'time', 10, 'latitude', 5, 'longitude', 5}, 'Datatype', 'double’);
```

Lecture 3. Exercise 1.

- Create a NetCDF file (‘example.nc’) with two variables (temperature, salinity). nccreate
- The variables are 3D, with dimensions for time, latitude, longitude.
- Write attributes to the variables, specifying variable units and long names. ncwriteatt
- Write a global attribute, specifying the exercise number and your name.
- View summary of created .nc file (ncdisp)
- Get info of created .nc file (ncinfo)

Add descriptions to your code and save the code.

Lecture 3. Exercise 2.

- Create a random variable and add it to the .nc variable. (rand, ncwrite)
- Load the same (entire) variable from the .nc file. (ncread)
- Load a subset of the same variable by defining the start and end indices of the dimensions:

  t = ncread(filename,’varname’,[xs,ys,zs],[xe,ye,ze])

- Create a subset of the loaded (entire) variable so that the dimensions will be (1,N,N). Make the variable more compact by using squeeze.

Lecture 3. Exercise 3.

- Create a condition, which checks if a file exists and deletes it, when it does exist. filename = “example2.nc”, exist, if-else-end
- Create .nc file

```
nc = netcdf.create(filename, 'NETCDF4’);
```

- Set dimension lengths for longitude (nx), latitude (ny), and time (nt). (Used when creating data variables)
- Define dimensions

```
dim_lon = netcdf.defDim(nc,'xaxis_1',nx);
dim_lat = netcdf.defDim(nc,'yaxis_1',ny);
```

- Define unlimited dimension

```
unlim_id = netcdf.defDim(nc, 'Time', netcdf.getConstant('NC_UNLIMITED'));
```

Lecture 3. Exercise 3 (continued).

- Define variables

```
varid1 = netcdf.defVar(nc, 'xaxis_1', 'double', dim_lon);
varid2 = netcdf.defVar(nc, 'yaxis_1', 'double', dim_lat);

varid5 = netcdf.defVar(nc, 'Time', 'double', []);

varid4 = netcdf.defVar(nc, 'temp', 'double', [dim_lon dim_lat unlim_id]);

netcdf.endDef(nc);
```

Lecture 3. Exercise 3 (continued).

- Define attributes

```
netcdf.putAtt(nc,varid1,'long_name','xaxis_1');
netcdf.putAtt(nc,varid1,'units','none');
netcdf.putAtt(nc,varid1,'cartesian_axis','X');

netcdf.putAtt(nc,varid2,'long_name','yaxis_1');
netcdf.putAtt(nc,varid2,'units','none');
netcdf.putAtt(nc,varid2,'cartesian_axis','Y');

netcdf.putAtt(nc,varid5,'long_name','Time');
netcdf.putAtt(nc,varid5,'units','time level');
netcdf.putAtt(nc,varid5,'cartesian_axis','T’);

netcdf.putAtt(nc,varid4,'long_name','temp');
netcdf.putAtt(nc,varid4,'units','none');
```

Lecture 3. Exercise 3 (continued).

- Put data into the variables x and y

```
netcdf.putVar(nc, varid1, 1:nx);
netcdf.putVar(nc, varid2, 1:ny);
netcdf.putVar(nc, varid5, 1);
```

- Create a random temperature variable

```
temperature = rand(nx,ny,nt);
```

- Define start indices (0-based in NetCDF)

```
start = [0, 0, 0]; % Start at the beginning of each dimension
```

- Define count (how many elements to write in each dimension)

```
count = [nx, ny, nt]; % Write all the data
```

- Correctly write the temperature data

```
netcdf.putVar(nc, varid4, start, count, temperature);
```

- Close .nc file

```
netcdf.close(nc)
```

Add descriptions to your code and save code.

Lecture 3. Exercise 4.

- Get file series_bot_lumi_057_a0.nc from Google drive.
- Get info on file.
- Visualize some parameters (make at least 2 different figures):
  - E.g. parameter change per time in one ‘rvk’ etc.
  - …
- Describe and save your code.